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Interactive GUI for Sinusoidal Gait Control and Real-Time Bipedal Robot Simulation in Webots

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ABSTRACT This paper presents an interactive graphical user interface (GUI) for real-time tuning of sinusoidal gait parameters for the Vocational Bipedal Robot (VoBiRo) in the Webots simulation environment. The proposed system eliminates the conventional trial-and-error workflow that requires repeated code modification, recompilation, and simulation restarts by employing a non-blocking UDP communication framework. This architecture enables continuous online adjustment of gait parameters without interrupting the high-frequency locomotion control loop, thereby significantly improving tuning efficiency. To validate the proposed framework, a One-Factor-at-a-Time (OFAT) optimization was conducted on five key gait parameters: period time, lateral swing amplitude, foot lift height, double support ratio (DSR), and arm swing gain. The optimized parameters were evaluated on a closed-loop oval trajectory requiring continuous heading changes, with locomotion stability quantified using the root mean square error (RMSE) of the center of mass (CoM) roll and pitch angles. Compared with the default configuration, the optimized gait reduced CoM roll RMSE by 12.4% and CoM pitch RMSE by 9.6%. Furthermore, Taguchi Signal-to-Noise (S/N) analysis with the Smaller-the-Better criterion confirmed improved robustness, yielding gains of 1.14 dB for lateral stability and 0.87 dB for sagittal stability. The proposed GUI provides an efficient and data-driven platform for rapid gait parameter optimization, facilitating intuitive tuning and improving the dynamic stability of bipedal robot locomotion in simulation.

INDEX TERMS Humanoid robots, bipedal locomotion, gait pattern generation, sinusoidal oscillators, parameter tuning, Webots simulation, walking stability metrics

I. INTRODUCTION

BIPEDAL robots designed to operate in human-centered environments must master stable walking behavior, which remains one of the most fundamental and challenging problems in robotics [1]–[4]. Stability in bipedal locomotion is critically important because walking performance directly affects the robot's ability to safely interact with humans, navigate complex environments, and execute various tasks in real-world applications. Unlike wheeled or quadrupedal platforms, bipedal robots must continuously coordinate body posture, Center of Mass (CoM) movement, foot placement, and multi-joint motion simultaneously to maintain balance while generating forward locomotion [5]–[7]. Even small deviations in posture or gait trajectory can lead to instability and falling, making balance control a primary concern in bipedal robot development [8], [9].

Bipedal robot locomotion is inherently dynamically unstable due to the continuously changing support conditions during walking. During single-leg support phases, the support polygon is reduced to the area of a single foot, significantly limiting the robot's stability margin and increasing sensitivity to disturbances. As a result, bipedal robots become highly vulnerable to external perturbations, uneven terrain, actuator inaccuracies, and inappropriate gait parameter configurations [10], [11]. Maintaining stable locomotion therefore requires precise synchronization between sensory feedback, dynamic modeling, and motor control systems [12].

Various gait generation methods have been developed for bipedal robots to achieve stable, efficient, and adaptive walking behavior. One of the most widely used approaches is the

Linear Inverted Pendulum Model (LIPM) [13], [14], which simplifies robot dynamics to facilitate Center of Mass (CoM) trajectory planning and balance control. Another commonly implemented approach is Zero Moment Point (ZMP)-based control [15]–[17], where walking stability is maintained by ensuring that the resultant moment remains within the robot's support polygon. In addition, sinusoidal oscillator-based gait generation has gained attention due to its simplicity, low computational requirements, and ability to generate smooth and periodic walking trajectories suitable for compact robotic platforms [18], [19]. More advanced methods include Central Pattern Generator (CPG)-based locomotion [20] inspired by biological neural systems, reinforcement learning approaches for adaptive motion generation [21], [22], and Model Predictive Control (MPC) [23]–[26] for dynamic balance optimization. Each gait generation method offers different advantages and limitations in terms of stability, computational complexity, adaptability, and implementation difficulty, making the selection of an appropriate approach highly dependent on the robot structure, control objectives, and operating environment [27]–[29].

Considering the highlighted advantages of the sinusoidal oscillator approach, this method is highly suitable for the locomotion system of the bipedal robot. Specifically, the application of the sinusoidal gait pattern on a bipedal robot utilizes mathematical sine waves to produce continuous and periodic joint trajectories, allowing for coordinated and rhythmic movements of the hip, knee, and ankle to achieve stable walking. In this paper, the Vocational Bipedal Robot (VoBiRo) is presented as the primary platform used in this research. The VoBiRo is a custom 22-degree-of-freedom (DoF) humanoid platform designed for locomotion and gait generation research, sharing kinematic complexities common in highly articulated bipedal hardware [5], [6]. To safely evaluate the proposed gait generation model before physical implementation, the walking behavior is first simulated and analyzed within the Webots environment [30]. This simulation approach enables the detailed observation of the robot's movement, Center of Mass (CoM) trajectories, and walking stability under various parameter configurations without risking hardware damage.

Despite its conceptual simplicity, the successful implementation of sinusoidal oscillators depends critically on the precise selection of gait parameters, such as oscillation frequency, step amplitude, and phase synchronization [31]–[34]. To address this challenge and support real-time monitoring, a Graphical User Interface (GUI) was developed using the Processing IDE [35]. This GUI enables interactive gait parameter tuning, real-time visualization of robot motion data, and direct communication with the robot control system [36], [37]. By integrating this interactive platform, the proposed framework simplifies the complex parameter tuning process and allows for an efficient and systematic evaluation of walking stability.

The novelty of this work lies in the development of an integrated GUI-assisted sinusoidal gait control framework for VoBiRo, combining real-time parameter tuning, locomotion monitoring, and quantitative stability evaluation within a unified simulation environment. Unlike existing studies that primarily emphasize gait generation algorithms, the proposed framework facilitates systematic analysis and interactive optimization of locomotion behavior through direct user interaction and real-time feedback. It is important to clarify the scope of this contribution: neither the sinusoidal-oscillator gait generator nor the One-Factor-At-A-Time tuning procedure is proposed as a novel algorithm. Both are established techniques adopted deliberately for their low computational complexity and interpretability. The contribution of this work is instead at the systems-integration level—a non-blocking UDP-based GUI architecture that eliminates the conventional code-recompilation and simulation-restart bottleneck, coupled with a reproducible, quantitative (RMSE and Taguchi S/N) workflow for evaluating gait-parameter changes in real time. This positions the study as an engineering and methodological contribution rather than a new locomotion-control or optimization algorithm. The contributions of this study are summarized as follows:

1) Development of a sinusoidal oscillator-based gait generation framework for the 22-DoF VoBiRo with low computational complexity and smooth periodic locomotion;
2) Design and implementation of a Processing-based graphical user interface (GUI) integrated with Webots via non-blocking UDP communication for real-time gait parameter adjustment and monitoring;
3) Demonstration of the GUI-assisted workflow through a systematic One-Factor-At-A-Time (OFAT) tuning case study of five critical gait parameters—Period Time, Lateral Swing Amplitude, Foot Lift Height, Double Support Ratio, and Arm Swing Gain—based on Root Mean Square Error (RMSE) of Center of Mass (CoM) stability, illustrating the framework's applicability rather than proposing OFAT as a novel optimization method;
4) Comparative validation of the Default and OFAT-Optimized configurations on a closed-loop oval reference track, together with statistical confirmation of the resulting stability improvements using the Taguchi Signal-to-Noise (S/N) ratio.

II. VOBIRO (VOCATIONAL BIPEDAL ROBOT)

The VoBiRo was developed as a 22-degree-of-freedom (22 DoF) bipedal robotic platform for research in humanoid locomotion, gait generation, and walking stability analysis. The robot configuration consists of four single-degree-of-freedom (DoF) joints on each arm, six single-DoF joints on each leg, and two single-DoF joints on the pan-tilt head mechanism, resulting in a total of 22 DoF as shown in Figure 1 .

The VoBiRo utilizes Dynamixel XL-430 servomotors for all 22 DoF joints due to their compact design, integrated control features, position feedback capability, and suitability for synchronized multi-joint robotic applications [38],

[39]. These actuators enable precise angular position control and smooth coordinated motion during walking experiments while supporting serial communication for centralized motion control. The overall VoBiRo design emphasizes lightweight construction and modular architecture to achieve stable locomotion with manageable computational and mechanical complexity [40]–[42], making it a suitable platform for studying bipedal walking algorithms, gait stability, and humanoid-inspired motion control in both simulation and real-world implementations.

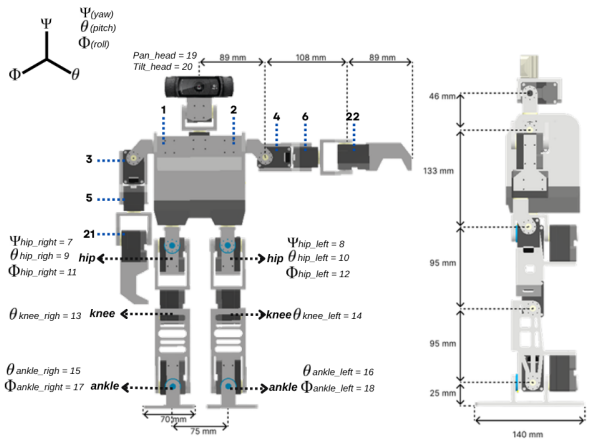


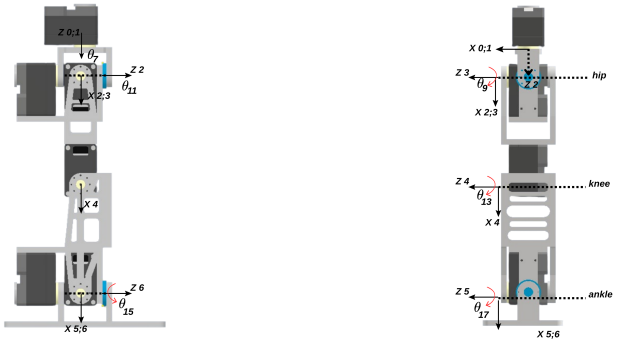
FIGURE 1: VoBiRo design showing servomotor placement and physical dimensions.

A. INVERSE KINEMATICS OF VOBIRO

In general, the Inverse Kinematics (IK) problem for a serial-chain robot is stated: given the pose (position and orientation) of the end frame of interest, calculate the joint values. On each leg, there are 6-DoF with the hip as the base and the ankle as the end effector. In this paper, the IK problem is solved using a trigonometric approach. For the right leg, the joints are symbolized by $(\theta_7, \theta_{11}, \theta_9, \theta_{13}, \theta_{17}, \theta_{15})$, corresponding to {yaw_hip, roll_hip, pitch_hip, pitch_knee, pitch_ankle, roll_ankle}. The upper thigh length is described by L_1 and the lower thigh by L_2. L_3 is the distance from the ankle to the shoe. Moreover, X and Y are the Cartesian coordinates of the foot placement when the robot is walking. Based on this geometric configuration, the analytical solutions to obtain the angle of each servomotor are formulated as shown in Equations (1) – (13).

For IK cases, we described the overview of the leg in three type views: top view, side view, and front view. The coordinate frames, joint placements, and directions of rotation for the 6-DoF right leg are illustrated in Figure 2. Further details of the foot placement from the top view are shown in Figure 3, while the precise joint angle definitions in the side and front views are depicted in Figure 4.

$$\theta_7 = heading_foot \tag{1}$$



(a) (b)

FIGURE 2: VoBiRo six-dof right leg coordinate frames: (a) Side view, (b) Front view.

$$R_0 = \sqrt{X^2 + Y^2} \tag{2}$$

$$\psi = \arctan\left(\frac{Y}{X}\right) - \theta_7 \tag{3}$$

$$sY = R_0 \sin\psi \tag{4}$$

$$sX = R_0 \cos\psi \tag{5}$$

$$R_2 = \sqrt{R_1^2 + sY^2} \tag{6}$$

$$\gamma = \arccos\left(\frac{L_1^2 + L_2^2 - R_2^2}{2L_1L_2}\right) \tag{7}$$

$$\beta = \arcsin\left(\frac{sX}{R_2}\right) \tag{8}$$

$$\alpha = \arcsin\left(\frac{L_2 \sin\gamma}{R_2}\right) \tag{9}$$

$$\theta_9 = \beta + \gamma \tag{10}$$

$$\theta_{13} = \pi - \alpha \tag{11}$$

$$\theta_{17} = \theta_9 - \theta_{13} \tag{12}$$

$$\theta_{11} = \theta_{15} \tag{13}$$

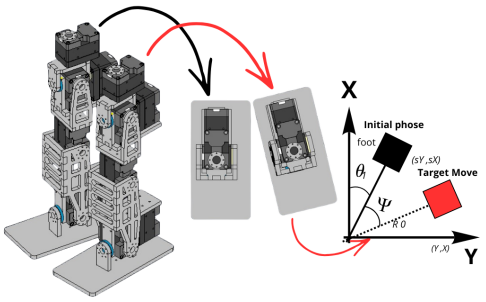


FIGURE 3: VoBiRo six-dof right leg top view.

B. GAIT PATTERN OF VOBIRO

1) Sinusoidal Gait Pattern

Sinusoidal Oscillator-Based Gait Pattern is a gait generation method for VoBiRo that utilizes sinusoidal functions

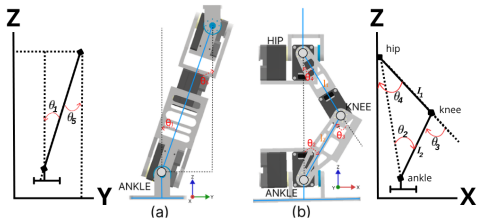


FIGURE 4: VoBiRo six-dof right leg: (a) Front View, (b) Side view.

to produce periodic and coordinated joint trajectories during walking motion [43]. In this approach, the movement of each joint, particularly on the hip, knee, and ankle, is generated using sinusoidal wave equations with specific amplitude, frequency, phase shift, and offset parameters. By adjusting these parameters, the robot can produce smooth and rhythmic walking patterns that resemble natural human locomotion. The sinusoidal oscillator method is widely used in bipedal robotics because of its simple mathematical formulation, low computational complexity, and ease of implementation in embedded robotic systems. The fundamental sinusoidal oscillator equation is given by:

$$\text{Output}(t) = A \cdot \sin\left(\frac{2\pi}{T} \cdot t - \varphi\right) + C, \qquad (14)$$

where A is amplitude, T is period, φ is phase shift, C is offset, and t is time. Parameters have intuitive physical meanings [31]. Amplitude directly controls step length (x-axis), foot clearance (z-axis), and lateral sway (y-axis). Period controls gait speed. Phase shift enables inter-limb coordination. Offset sets baseline posture.

Compared to more complex approaches such as the LIPM or ZMP-based control, the sinusoidal oscillator approach offers a simpler implementation with lower computational requirements .Figure 5 illustrates the block diagram of the Sinusoidal Oscillators as Gait Pattern framework for bipedal robots. The proposed system is divided into several interconnected modules that work sequentially to generate periodic walking trajectories, coordinate locomotion phases, and produce stable walking motion for the robot.

1) The first section is the Gait Parameters (User Input) module, which functions as the main configuration stage for gait generation. In this module, several walking parameters are defined, including oscillation amplitude, walking frequency, phase shift between the left and right legs, angular offset, step period, and gait type. These parameters determine the characteristics of the walking motion such as stride length, walking speed, and movement rhythm. The configured parameters are then transmitted to the sinusoidal oscillator module for trajectory generation.
2) The second section is the Sinusoidal Oscillators (Joint Generators) module. This module is responsible for generating periodic joint trajectories for the hip, knee, and ankle joints of both the left and right legs using sinusoidal equations. Each joint trajectory is represented using a sinusoidal function to create smooth and rhythmic locomotion patterns. The generated trajectories define the desired joint angles for the robot during walking cycles. Synchronization between both legs is achieved by introducing phase differences between the oscillators, enabling alternating swing and stance motions during locomotion.
3) The third section is the Phase Scheduler and Coordination module, which manages the walking phase transitions and synchronization between the left and right legs. This module controls the gait state machine, including swing phase, stance phase, step switching, and double support conditions. In addition, the central phase generator regulates the timing and phase updates required to maintain continuous and coordinated walking motion. Proper coordination within this module is essential to ensure stable and balanced locomotion.
4) The fourth section is the Inverse Kinematics and Trajectory Mapping module. In this stage, the generated sinusoidal trajectories are mapped into the robot joint space to produce executable joint commands. If foot trajectory control is implemented, inverse kinematics calculations are used to determine the corresponding joint angles required to achieve the desired foot positions. The generated trajectories may also undergo smoothing and interpolation processes to reduce abrupt motion changes and improve walking smoothness. The final output of this module is the joint command vector used to control the robot actuators.
5) The fifth section is the Outputs module, which sends the generated joint commands to the robot or simulation environment. The output may consist of joint angle references or desired torque commands depending on the control implementation. In addition, this section provides visualization outputs such as joint angle plots, gait phase visualization, robot animation, and foot trajectory monitoring to support locomotion analysis and system evaluation.
6) The final section is the Feedback Module, which receives measured data from the robot or simulation environment. The feedback data include joint positions, inertial measurement unit (IMU) data, Center of Mass (CoM) information, foot sensor readings, and gait contact states. These measured values are compared with the desired joint trajectories to calculate tracking errors and evaluate locomotion performance. The feedback information is then used to update the gait generation process and improve walking stability and trajectory tracking accuracy.

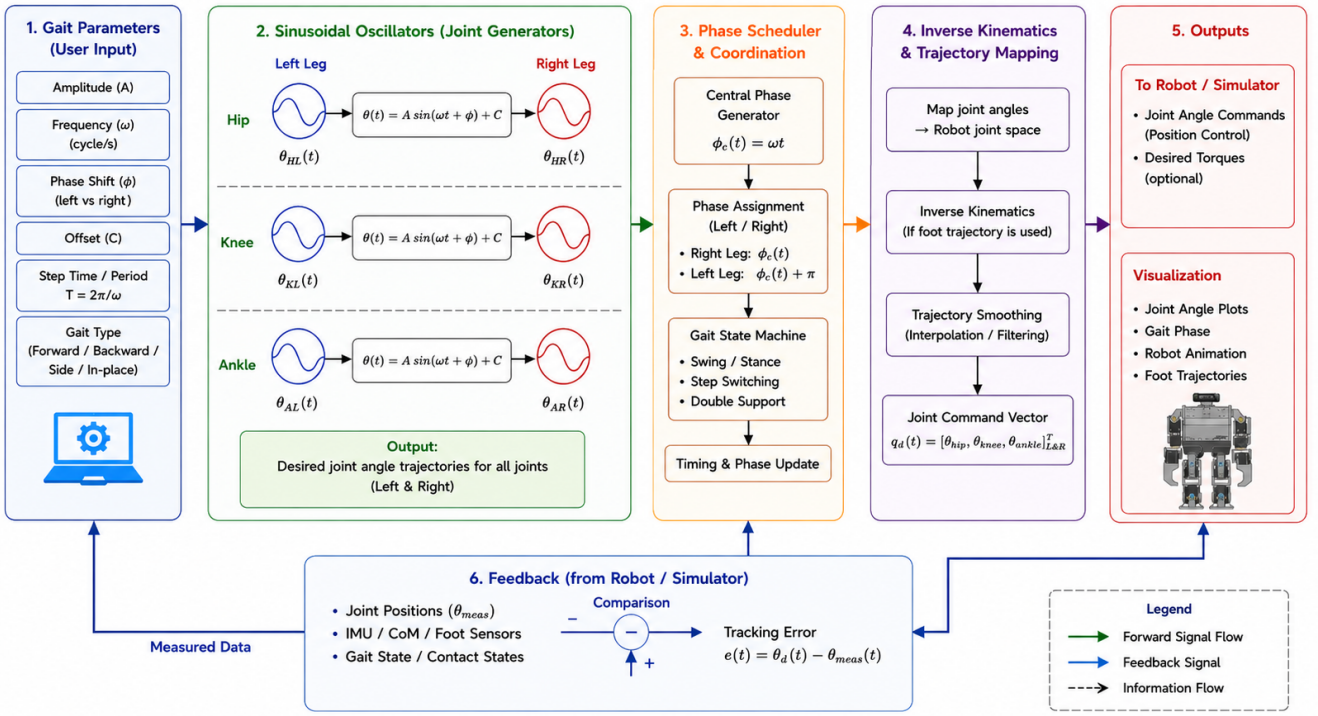


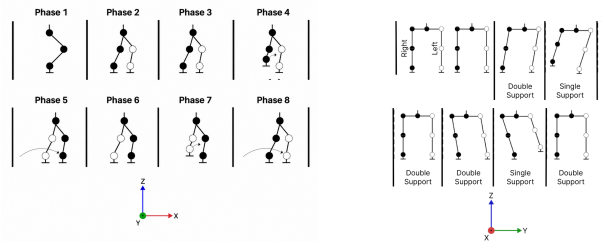
FIGURE 5: Block Diagram of the Sinusoidal Oscillators as Gait Pattern framework for VoBiRo

2) Gait Decomposition for Bipedal Robot

Walking is decomposed into eight phases to structure trajectory generation [44], [45]:

1) **Phase 1:** Initialization — Define coordinate system
2) **Phase 2:** Preparation — Position feet for walking initiation
3) **Phase 3:** CoM shift to left leg — Bilateral support, CoM transition
4) **Phase 4:** Right leg swing — Right foot lift and forward motion
5) **Phase 5:** Right leg landing — Right foot contact, CoM preparation
6) **Phase 6:** CoM shift to right leg — Bilateral support, CoM transition
7) **Phase 7:** Left leg swing — Left foot lift and forward motion
8) **Phase 8:** Left leg landing — Left foot contact, CoM preparation

This decomposition clarifies stability mechanisms as seen in Figures 6a and 6b. Phases 3 and 6 establish lateral CoM shift during bilateral support, positioning CoM over the stance leg before swing initiation. Phases 4 and 7 execute swing motion with controlled foot clearance. Phases 5 and 8 complete the transition, preparing for the next stance shift. This coordination prevents the fundamental instability of bipedal walking by proactively shifting CoM during stable bilateral support [46].



(a) (b)

FIGURE 6: Eight-phase gait cycle: (a) Sagittal plane, (b) Frontal plane.

The hip and ankle trajectory generation during a single gait cycle is illustrated in Figure 7, which shows both the sagittal plane components for forward and vertical motion and the frontal plane components for lateral weight transfer. The circles in the diagram represent joint motion at different time points within the gait cycle, demonstrating the smooth sinusoidal-based trajectories that guide the robot's coordinated movement.

C. WEBOTS SIMULATION OF VOBIRO

In this paper, the CAD model of the VoBiRo converted and integrated into the Webots environment to develop a realistic simulation model for gait generation and walking stability analysis. The conversion process preserved the robot's

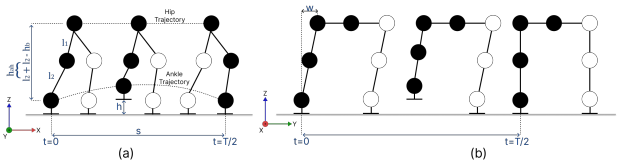


FIGURE 7: Joint trajectories over a single gait cycle: (a) sagittal Forward/vertical motions, (b) Frontal lateral CoM sway.

original 22 DoF kinematic structure, body dimensions, joint configurations, and mass distribution while adding actuator, sensor, and joint constraint definitions for physics-based simulation. Webots was selected over alternatives such as Gazebo or MuJoCo because it natively supports direct conversion of CAD assemblies into simulation models with preserved kinematic structure and mass distribution, and it exposes built-in sensor emulation (IMU, foot contact) together with external-controller communication (e.g., UDP) without requiring additional middleware, which simplified integration with the non-blocking GUI framework proposed in this work.

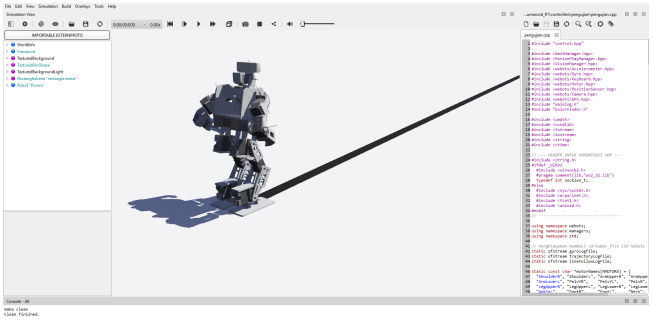


FIGURE 8: Webots Simulation of the VoBiRo

As shown in Figure 8, this approach enables efficient testing and evaluation of locomotion algorithms in a virtual environment before real-world implementation, reducing hardware risks and supporting detailed analysis of walking performance and CoM stability.

III. GUI-BASED SINUSOIDAL GAIT CONTROL FOR BIPEDAL ROBOTS

The development of a Graphical User Interface (GUI)-based sinusoidal gait control system plays a crucial role in simplifying parameter configuration, real-time monitoring, and performance evaluation for bipedal robots. By integrating a sinusoidal oscillator-based gait generator with an interactive GUI platform, the proposed framework enables users to efficiently adjust key walking parameters—such as oscillation amplitude, frequency, phase shift, and joint offsets—during locomotion experiments. In this study, the GUI was developed using the Processing IDE to facilitate real-time 3D visualization and interactive control. To ensure low-latency data transmission without interrupting the robot's main control loop, communication between the Processing GUI and the robot control system (Webots) is established via a non-blocking User Datagram Protocol (UDP), as illustrated in Figure 9. This integrated approach significantly improves the flexibility of gait tuning and provides a more effective environment for analyzing walking stability and locomotion behavior. In the present study, the Processing GUI and the Webots controller communicate over UDP on the same local workstation; because the link does not traverse a wireless network, transmission latency is expected to be negligible relative to the 500–600 ms gait-cycle period used in this work. We did not, however, formally instrument round-trip latency or evaluate the effect of packet loss in this study. Both become materially more important once the framework is deployed over a wireless link to physical hardware, and we identify UDP latency/packet-loss characterization under such conditions as a necessary step for future real-robot validation.

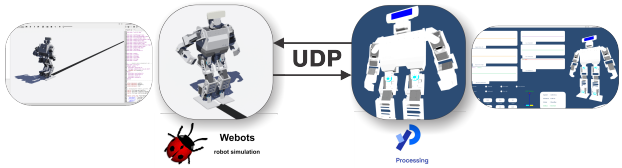


FIGURE 9: System architecture illustrating the non-blocking User Datagram Protocol (UDP) communication between the Webots simulation and the Processing GUI.

A. GRAPHICAL USER INTERFACE DASHBOARD

To support systematic parameter evaluation, a GUI-based monitoring and data acquisition system was developed using the Processing IDE, as shown in Figure 10. The system enables simultaneous real-time visualization of eight data channels: per-joint motor angles of the left and right legs, IMU roll and pitch readings, lateral CoM displacement (Y-axis), vertical CoM displacement (Z-axis), foot trajectory (Z-axis), and a 2D CoM map. This multi-channel monitoring capability allows immediate observation of gait instability patterns—such as excessive lateral sway, foot dragging, or asymmetric joint motion—during parameter variation experiments without interrupting the walking cycle. All kinematic and sensor data are logged continuously at 100 Hz and exported to structured spreadsheet files, providing the quantitative dataset used for the performance analysis in Section IV. The interactive parameter panel further allows on-the-fly adjustment of gait parameters, including period time, foot lift height, lateral swing amplitude, and positional offsets, enabling rapid iterative tuning between experimental runs. Gait-parameter updates are transmitted asynchronously via UDP and are decoupled from the underlying Webots physics update loop. All trials in this study were conducted on flat, obstacle-free reference tracks (Figs. 11 and 18) without externally applied disturbances (e.g., pushes or uneven terrain); evaluating disturbance rejection and push-recovery is identified as future work (Section V).

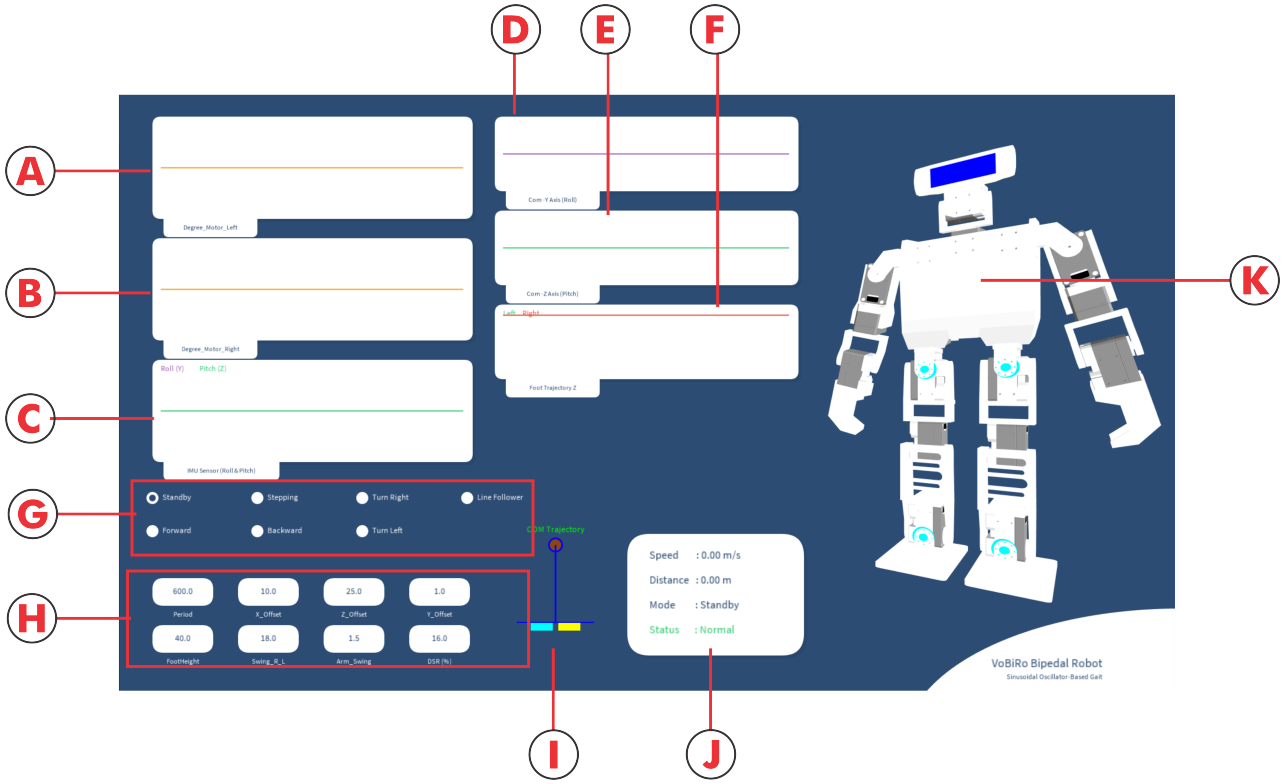


FIGURE 10: The developed GUI dashboard, featuring real-time 3D kinematic visualization, live data plotting of sensor responses, and an interactive control panel for sinusoidal gait parameter tuning.

TABLE 1: Description of GUI Dashboard Components (refer to Fig. 10)

No.	Lbl	Component	Description
1	**A**	Left Leg Motor Angles	Real-time plot of the kinematic joint trajectories (hip, knee, ankle) for the left leg.
2	**B**	Right Leg Motor Angles	Real-time plot of the kinematic joint trajectories (hip, knee, ankle) for the right leg.
3	**C**	IMU Sensor Graph	Plots the robot's angular orientation (Roll and Pitch) based on simulated IMU feedback.
4	**D**	CoM X-Axis (Roll)	Monitors lateral sway of the Center of Mass to detect side-to-side instabilities.
5	**E**	CoM Y-Axis (Pitch)	Monitors vertical/sagittal displacement of the Center of Mass to prevent falls.
6	**F**	Foot Trajectory (Z-Axis)	Tracks vertical ground clearance of both feet to ensure proper lifting during swing phase.
7	**G**	Locomotion Mode Selector	Radio buttons to command the robot's active state (Standby, Stepping, Forward, etc.).
8	**H**	Interactive Parameter Panel	Input capsules allowing users to adjust sinusoidal gait parameters on-the-fly via UDP.
9	**I**	2D CoM Visualizer	Graphical pendulum representation showing real-time CoM sway relative to foot placement.
10	**J**	Status & Perf. Dashboard	Displays live calculated metrics including speed, distance, mode, and fall detection status.
11	**K**	3D Kinematic Visualizer	Real-time 3D rendering of the VoBiRo robot for immediate visual posture feedback.

The functional layout of the developed GUI is organized into specific monitoring and control zones, denoted by labels A through J in Figure 10. A detailed description of each component's function is provided in Table 1.

B. TUNING PARAMETERS OF VOBIRO

Through the developed GUI, various locomotion parameters can be configured and transmitted to the robot's control system in real time. Before identifying the most effective walking pattern, it is necessary to establish a reliable baseline configuration. To this end, ten fundamental gait parameters were defined, each serving a specific kinematic function in generating a stable bipedal walk. Table 2 summarizes these parameters, their units, and their functional role within the sinusoidal oscillator framework described in Section II-B1. These parameter definitions form the basis of the One-Factor-At-A-Time (OFAT) tuning procedure presented in Section IV. Of these ten parameters, the five selected for OFAT tuning—Period Time, Lateral Swing Amplitude, Foot Lift Height, Double Support Ratio, and Arm Swing Gain—were chosen because they directly govern gait-cycle timing, lateral weight transfer, foot ground-clearance, ground-contact duration, and yaw-momentum compensation, i.e., the kinematic factors most directly coupled to the CoM roll/pitch dynamics used as the stability metric in this study. The remaining five parameters (X/Y/Z Offset, Pitch & Hip Pitch Offset, Pelvis Offset, Forward/Back Ratio, and Swing Top-Down Factor) primarily shape static posture and trajectory profile rather than dynamic CoM balance and were therefore held at the baseline values listed in Table 3 throughout this study. Extending the tuned set to the full ten-parameter space is identified as future work.

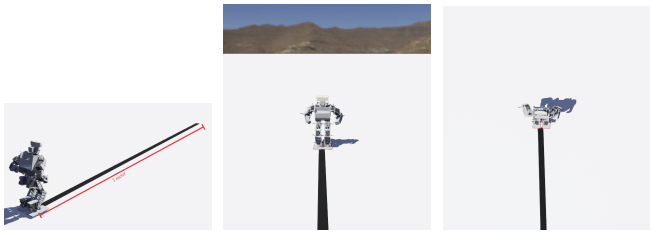
TABLE 2: Gait Tuning Parameters, Units, and Functions

No.	Parameter	Unit	Function
1	Period Time	ms	Total duration of one complete gait cycle
2	Lateral Swing Amplitude	mm	Maximum lateral displacement of the CoM
3	Foot Lift Height	mm	Maximum vertical clearance of the swing foot
4	Double Support Ratio (DSR)	%	% of gait cycle where both feet touch the ground
5	Arm Swing Gain	–	Multiplier for arm amplitude to cancel yaw momentum
6	X, Y, Z Offset	mm	Initial spatial displacement of the CoM
7	Pitch & Hip Pitch Offset	deg	Forward/backward tilt angle of trunk and hips
8	Pelvis Offset	mm	Positional adjustment of the pelvis for hip sway
9	Forward/Back Ratio	–	Temporal asymmetry between swing and stance phases
10	Swing Top-Down Factor	–	Shape factor for the vertical foot-trajectory profile

C. EXPERIMENTAL PROTOCOL

To systematically evaluate the performance of the proposed sinusoidal gait generation framework, a comprehensive experimental protocol was established within the Webots simulation environment. All testing scenarios were executed on a designated linear reference track to provide an objective and standardized measure of locomotion quality across different parameter sets.

A key aspect of this experimental setup is the implementation of the *Line Follower* mode, as illustrated in Figure 11. Throughout the entire testing process, the robot operated autonomously using its vision-based system to follow a straight black reference line. This ensures that the robot maintains a consistent heading and stays within the designated track. By using this mode, any observed instabilities, lateral deviations, or falls can be attributed solely to variations in the gait parameters rather than to errors in manual steering or direction control.



(a) (b) (c)

FIGURE 11: The simulated linear reference track used for the line-follower walking experiments in Webots, showing (a) perspective, (b) front, and (c) top views of the VoBiRo.

The testing procedure is categorized into four distinct validation scenarios designed to assess different aspects of the robot's performance:

1) **Gait Parameter Sensitivity Analysis**: evaluates how the robot's locomotion characteristics respond to systematic changes in control inputs while maintaining its path in line follower mode, observing the direct impact on step length, walking speed, and overall balancing behavior.
2) **Walking Stability Evaluation**: quantifies dynamic stability margins by subjecting the robot to continuous walking trials, recording the number of successful steps completed before failure and the spatial trajectories of the CoM to detect instability patterns.
3) **Real-Time Tuning Validation**: assesses the system's responsiveness and transition smoothness when gait parameters are modified on-the-fly via the Processing GUI dashboard while the robot is actively walking, ensuring that UDP communication does not interrupt the locomotion logic.
4) **Trajectory and Joint Angle Verification**: validates kinematic tracking accuracy by logging the actual angular positions of the hip, knee, and ankle joints and cross-examining them with the analytical sinusoidal reference signals.

IV. RESULTS AND DISCUSSION

This section presents the simulation results obtained from the Webots environment to evaluate the proposed sinusoidal oscillator-based gait generation framework and the developed interactive GUI. The discussion is structured to reflect a systematic optimization process, beginning with the baseline configuration and progressing through a sequential parameter-tuning strategy.

A. EXPERIMENTAL SETUP AND BASELINE CONFIGURATION

The experimental validation was initiated by establishing a stable baseline configuration derived from VoBiRo's physical constraints. Operating under these default conditions ensures that the robot can perform basic forward locomotion in Line Follower mode before any optimization takes place. The default gait parameters are detailed in Table 3. The tested range for each parameter in the OFAT sweeps below (Section IV-B) was bounded by VoBiRo's physical and kinematic constraints (leg dimensions, joint travel, and actuator torque limits) and refined through preliminary pilot trials to exclude ranges that produced immediate falls prior to the reported sweeps. CoM roll and pitch RMSE were selected as the primary stability indicators because they directly correspond to VoBiRo's two dominant instability modes—lateral tip-over and sagittal (forward/backward) falls; other potentially informative metrics, such as yaw drift, step-length variance, and Zero-Moment-Point margin, were not evaluated in the present study and are identified as directions for future evaluation (see Section V).

TABLE 3: Default Gait Parameters for Baseline Configuration

No.	Parameter	Value
1	Period Time (T)	500 ms
2	Lateral Swing Amplitude (w)	18 mm
3	Foot Lift Height (h)	40 mm
4	Double Support Ratio (DSR)	10%
5	Arm Swing Gain	1.5
6	X, Y, Z Offset	10, 1, 25 mm
7	Pitch & Hip Pitch Offset	$-5°, 5°$
8	Pelvis Offset	3 mm
9	Forward/Back Ratio	0.5
10	Swing Top-Down Factor	5.0

To systematically isolate and identify the most effective walking pattern without computational bottlenecks, the developed UDP-based interactive GUI was used to conduct a One-Factor-At-A-Time (OFAT) parameter-tuning procedure. In this approach, one specific parameter is optimized in real time while the others are held constant. Once the best value for a parameter is empirically identified—...characterized by the lowest Root Mean Square Error (RMSE) for Center of Mass (CoM) stability over an 11-second observation window—it is retained and applied to the subsequent stages. OFAT was adopted for this study because it allows each parameter's effect on CoM RMSE to be observed directly and interactively through the GUI, which was the primary capability under evaluation. We note that OFAT does not account for interaction effects between parameters and can converge to a locally, rather than globally, optimal configuration; adopting interaction-aware, multivariable optimization methods (e.g., Bayesian optimization, genetic algorithms, particle swarm optimization, CMA-ES, or Taguchi orthogonal-array design) is identified as future work (Section V).

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2}. \tag{15}$$

In Eq. 15, x_i is the IMU-measured CoM roll or pitch angle (rad) at sample i, and $\hat{x}_i$ is the corresponding reference value, taken as the nominal upright/level equilibrium orientation (i.e., zero tilt); the RMSE therefore quantifies deviation from level CoM orientation. Roll is defined as rotation about the robot's forward (sagittal, X) axis, representing lateral tilt, and Pitch as rotation about the lateral (frontal, Y) axis, representing forward/backward tilt, consistent with the coordinate convention in Fig. 1. Data were sampled at the GUI's 100 Hz logging rate. The 11-second observation window corresponds to one full traversal of the reference track—the straight 3 m line used in Sections IV-A/B, or one lap of the closed-loop oval track used in Sections IV-C/D—at VoBiRo's nominal walking speed, spanning several complete gait cycles under identical conditions across all compared configurations. All RMSE values reported in Tables 4–11 correspond to a single 11-second trial per configuration ($n = 1$), executed under fixed initial conditions and Webots' default deterministic physics settings without injected randomness; standard deviation and confidence intervals could therefore not be computed from the present dataset, and reporting variability measures from repeated, randomized-condition trials is identified as necessary future work.

The systematic narrative follows this progressive sequence: (1) Period Time, (2) Lateral Swing Amplitude, (3) Foot Lift Height, (4) Double Support Ratio, and (5) Arm Swing Gain.

1) Real-Time Tuning Setup via Interactive GUI

To facilitate the real-time OFAT tuning process, a dual-display workstation was configured, as depicted in Fig. 12. The primary control interface, developed using Processing, operates on one screen to allow continuous, on-the-fly adjustments of the sinusoidal gait parameters via UDP commands. Simultaneously, the Webots physics engine runs on an adjacent monitor, providing immediate 3D visual feedback of the VoBiRo's locomotion and dynamic stability in response to the injected parameters. This interactive setup eliminates the traditional need for repeated code recompilation, drastically reducing the heuristic tuning cycle time.

B. SYSTEMATIC GAIT PARAMETER OPTIMIZATION VIA INTERACTIVE GUI

1) Finding the Optimal Period Time

The optimization begins by varying the total gait cycle duration (e.g., from 400 ms to 1000 ms) from the default setting, in order to find the resonant stepping rhythm that balances forward momentum without overwhelming the robot's inertia. The period time is a foundational parameter; setting it incorrectly physically limits the effectiveness of any subsequent tuning. When the period was set too slowly (1000 ms and 800 ms), prolonged single-support phases caused the locomotion trajectory to terminate prematurely due to a severe

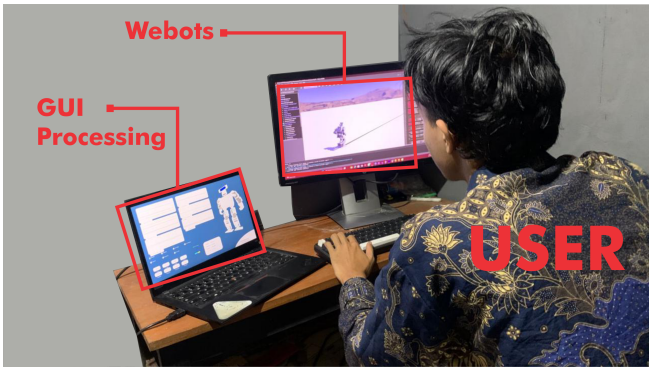
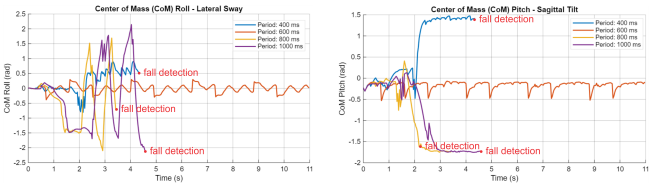


FIGURE 12: The experimental workstation setup for real-time gait tuning, demonstrating the simultaneous operation of the interactive Processing GUI and the Webots 3D simulation environment. A demonstration video of the proposed system is available at https://youtu.be/3t1CpGPR8Io.

forward fall, indicated by large fluctuations along the CoM Pitch axis. Conversely, an aggressively short period of 400 ms induced excessive high-frequency inertial forces that the actuators could not accurately track.



(a) (b)

FIGURE 13: Kinematic responses under varying Period Times: (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

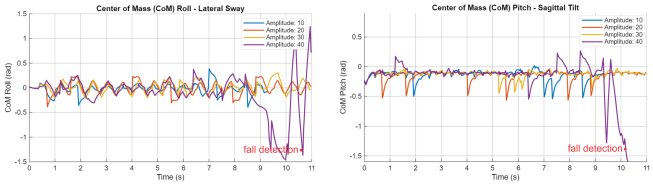
As illustrated in Figure 13 and quantitatively validated in Table 4, the 600 ms configuration was identified as the optimal period. This specific temporal duration allowed the robot to establish a stable limit cycle, yielding the lowest tracking errors across the evaluated spectrum (Roll RMSE: 0.1310 rad; Pitch RMSE: 0.1891 rad). By securing this resonant frequency first, the system established a stable locomotion rhythm without sacrificing the sagittal or lateral balance required for the upcoming spatial parameter adjustments.

TABLE 4: RMSE Results for Period Time Variations

No.	Condition	Roll (rad)	Pitch (rad)
1	400 ms	0.4341	0.9958
2	**600 ms**	**0.1310**	**0.1891**
3	800 ms	0.9692	1.0987
4	1000 ms	1.1137	1.1681

2) Finding the Optimal Lateral Swing Amplitude

Using the previously identified optimal period time ($T = 600$ ms), the lateral displacement of the CoM was adjusted via the GUI. This step seeks to define the precise hip-sway amplitude required to maintain a stable support polygon during the single-leg stance phase. Biomechanically, insufficient lateral swing prevents the robot from shifting its weight entirely over the supporting foot, predictably leading to lateral tip-overs. In contrast, an excessive lateral amplitude generates unwanted centrifugal forces that violently disrupt the robot's sagittal momentum and forward trajectory.



(a) (b)

FIGURE 14: Kinematic responses under varying Lateral Swing Amplitudes: (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

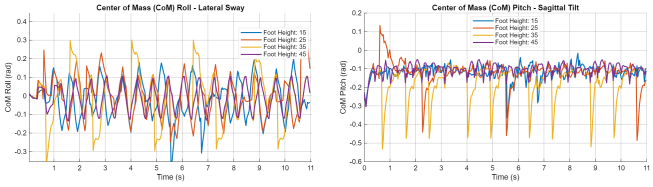
As shown in Figure 14 and detailed in Table 5, tuning the amplitudes between 10 mm and 40 mm revealed distinct behavioral boundaries. Amplitudes of 20 mm and 30 mm produced the most consistent CoM Roll behavior, while pushing the amplitude to 40 mm caused a sharp divergence and eventual failure after approximately 9 seconds. By monitoring the real-time CoM tracking errors, an amplitude of **30 mm** was empirically selected as the optimal lateral swing. This configuration recorded highly stable metrics (Roll RMSE: 0.1143 rad; Pitch RMSE: 0.1431 rad), ensuring a secure lateral weight transfer for the subsequent tuning stages.

TABLE 5: RMSE Results for Lateral Swing Amplitude Variations

No.	Condition	Roll (rad)	Pitch (rad)
1	10 mm	0.1080	0.1636
2	20 mm	0.1234	0.1748
3	**30 mm**	**0.1143**	**0.1431**
4	40 mm	0.4429	0.6450

3) Finding the Optimal Foot Lift Height

With the period and lateral swing optimized, the vertical clearance of the swing foot was tuned across the range of 15 mm to 45 mm. The goal is to determine the minimum lifting height necessary to prevent ground-scuffing while avoiding excessive vertical CoM bobbing that wastes energy. As detailed in Table 6 and Fig. 15, the 35 mm configuration exhibited a notable spike in Sagittal Tilt (Pitch) RMSE, as the exaggerated lifting motion destabilized the robot's forward posture. The 45 mm configuration yielded the lowest overall tracking errors (Roll RMSE: 0.0694 rad; Pitch RMSE: 0.1091 rad) and is therefore identified as the most dynamically stable clearance among the four tested conditions.



(a) (b)

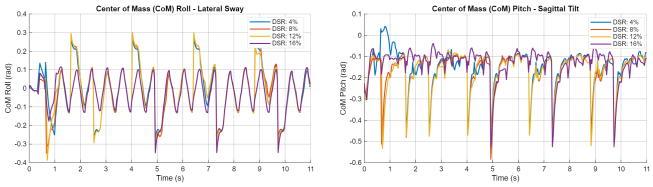
FIGURE 15: Kinematic responses under varying Foot Lift Heights: (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

TABLE 6: RMSE Results for Foot Lift Height Variations

No.	Condition	Roll (rad)	Pitch (rad)
1	15 mm	0.1023	0.1251
2	25 mm	0.1076	0.1417
3	35 mm	0.1316	0.1901
4	**45 mm**	**0.0694**	**0.1091**

4) Finding the Optimal Double Support Ratio (DSR)

Building on the previously optimized values, the Double Support Ratio (DSR) was adjusted within the range of 4% to 16%. DSR dictates the percentage of the gait cycle during which both feet simultaneously touch the ground, which is fundamental for absorbing impact transients. As shown in Table 7, an inadequately low DSR (4%) resulted in noticeably larger lateral and sagittal tracking errors, indicating insufficient time for weight transfer between legs. Within the tested range, a DSR of 16% achieved the lowest roll error (0.1028 rad) and pitch error (0.1511 rad), reflecting a favorable balance between fluid weight transfer and overall postural stability.



(a) (b)

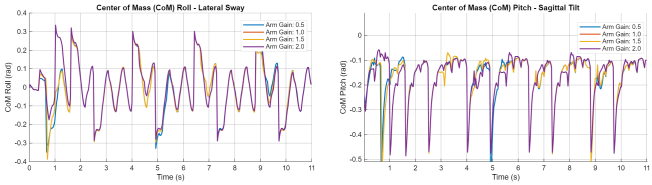
FIGURE 16: Kinematic responses under varying Double Support Ratios (DSR): (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

TABLE 7: RMSE Results for Double Support Ratio (DSR) Variations

No.	Condition	Roll (rad)	Pitch (rad)
1	4%	0.1262	0.1720
2	8%	0.1364	0.1912
3	12%	0.1316	0.1901
4	**16%**	**0.1028**	**0.1511**

5) Finding the Optimal Arm Swing Gain

Finally, the reciprocal arm-swing amplitude was fine-tuned to counteract the residual yaw momentum generated by the swinging legs. The gain was systematically varied from 0.5 to 2.0. As detailed in Table 8, increasing the gain to 2.0 effectively neutralized the leg-induced yaw and stabilized the trunk, yielding the lowest Pitch RMSE (0.1866 rad), with only a marginal increase in Roll RMSE (0.1367 rad) relative to the 1.0–1.5 configurations. The ability to tune this parameter on-the-fly highlights the practical utility of the proposed GUI in resolving complex multi-axis dynamic imbalances.



(a) (b)

FIGURE 17: Kinematic responses under varying Arm Swing Gains: (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

TABLE 8: RMSE Results for Arm Swing Gain Variations

No.	Condition	Roll (rad)	Pitch (rad)
1	0.5	0.1364	0.1912
2	1.0	0.1316	0.1901
3	1.5	0.1316	0.1901
4	**2.0**	0.1367	**0.1866**

C. COMPARATIVE EVALUATION OF DEFAULT AND OFAT-OPTIMIZED CONFIGURATIONS

After sequentially identifying the optimal value for each of the five gait parameters through the OFAT procedure (Sections IV-B1–IV-B5), the individually optimized values were combined into a single, unified configuration (600 ms period, 30 mm lateral swing, 45 mm foot lift, 16% DSR, and 2.0 arm swing gain). To clearly illustrate the parameter shifts resulting from the tuning process, Table 9 summarizes the exact values of the Default baseline versus the OFAT-Optimized configuration.

TABLE 9: Comparison of Gait Parameters: Default vs. OFAT-Optimized Configuration

Parameter	Unit	Default	OFAT-Optimized
Period Time (T)	ms	500	**600**
Lateral Swing Amplitude (w)	mm	18	**30**
Foot Lift Height (h)	mm	40	**45**
Double Support Ratio (DSR)	%	10	**16**
Arm Swing Gain	–	1.5	**2.0**

To verify that this combined configuration produces a measurable overall improvement under more demanding locomo-

tion conditions, the Default and OFAT-Optimized configurations were directly compared on a closed-loop oval reference track on Figure 18, which, unlike the straight 3 m line used in the initial sensitivity tests, requires the robot to sustain continuous heading changes while walking. Both configurations were evaluated over the same 11-second observation window under identical Line Follower conditions.

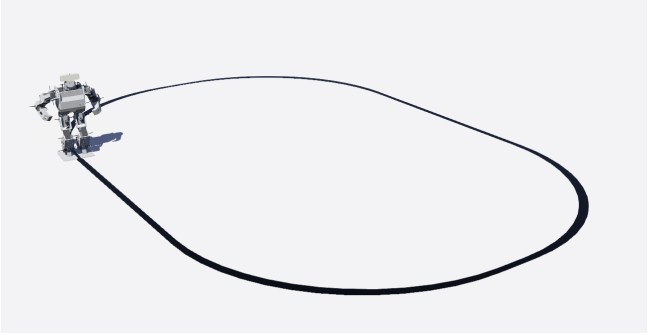
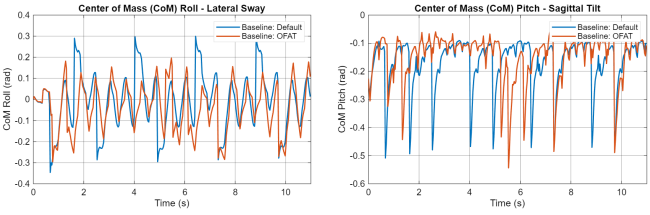


FIGURE 18: The closed-loop oval reference track used for the Default vs. OFAT-Optimized comparative validation.

Figure 19 presents the resulting kinematic responses of both configurations, while Table 10 summarizes the corresponding RMSE values for CoM Roll and CoM Pitch.



(a) (b)

FIGURE 19: Kinematic responses of the Default versus OFAT-Optimized configurations on the oval reference track: (a) CoM Roll representing lateral sway, (b) CoM Pitch representing sagittal tilt.

TABLE 10: RMSE Comparison Between Default and OFAT-Optimized Configurations (Oval Track)

Configuration	Roll (rad)	Pitch (rad)
Default Configuration	0.1343	0.1894
OFAT-Optimized Configuration	**0.1177**	**0.1713**

The lateral and sagittal balance metrics show a clear and consistent improvement under the OFAT-Optimized configuration. As illustrated in Figure 19a, the Default configuration exhibits markedly larger CoM Roll excursions during the first two gait cycles, peaking near $0.30\,\text{rad}$ and $-0.35\,\text{rad}$, before settling into a somewhat irregular oscillation pattern. The OFAT-Optimized configuration, while still oscillatory in nature due to the continuous turning demand of the oval path, maintains noticeably tighter peak amplitudes after the initial transient, yielding a 12.4% reduction in Roll RMSE (from $0.1343\,\text{rad}$ to $0.1177\,\text{rad}$).

A similar trend is observed for the sagittal CoM Pitch response (Figure 19b). The Default configuration produces deep, periodic forward-tilt troughs reaching as low as $-0.55\,\text{rad}$, indicating substantial sagittal destabilization at each footfall. The OFAT-Optimized configuration suppresses these extreme troughs to approximately -0.30 to $-0.35\,\text{rad}$, resulting in a 9.6% reduction in Pitch RMSE (from $0.1894\,\text{rad}$ to $0.1713\,\text{rad}$).

Taken together, these results indicate that the cumulative effect of the five individually-tuned OFAT parameters translates into a tangible overall stability improvement under more challenging, continuously turning locomotion conditions. While these RMSE reductions are descriptively meaningful, a formal statistical measure is needed to assess whether the observed improvement reflects a genuine reduction in dynamic disturbance sensitivity rather than ordinary trial-to-trial variation. This statistical validation, based on the same comparative dataset, is presented in the following subsection using the Taguchi Signal-to-Noise (S/N) ratio.

We emphasize that the RMSE and Taguchi S/N comparisons quantify the relative improvement of the OFAT-Optimized configuration over the Default configuration under matched, deterministic simulation conditions ($n = 1$ per configuration); they do not, by themselves, establish statistical robustness across repeated or randomized trials, nor do they validate performance under a jointly optimized, interaction-aware parameter search. Strengthening statistical validation through repeated randomized trials and adopting an interaction-aware multivariable optimization strategy are identified as priority future work (Section V).

Relative to CPG-based, ZMP-based, MPC-based, and reinforcement-learning gait generation, the sinusoidal-oscillator approach used here trades adaptability and formal balance guarantees for computational simplicity and interpretability: closed-form joint trajectories (Eq. 14) can be tuned and re-evaluated within a single gait cycle via the GUI, with O(1) per-joint computational cost and no online optimization solve required during walking, unlike the model identification (ZMP/MPC), training (RL), or oscillator-coupling design (CPG) overhead these methods typically require. This makes the proposed framework well suited to rapid, interactive exploration of gait behavior, but it does not offer the online disturbance-rejection or adaptability guarantees associated with ZMP/MPC or learned controllers; a quantitative, same-platform benchmark against these alternative paradigms was outside the scope of this study and is identified as future work. In its current form, the GUI supports manual, operator-driven parameter adjustment rather than automatic or closed-loop optimization; extending the interactive parameter panel to drive an automated search (e.g., Bayesian optimization or metaheuristic search) over the UDP interface is discussed further in Section V.

Extending the GUI and UDP communication architecture to higher-DoF humanoid platforms is expected to be archi-

tecturally straightforward, since each additional joint requires only an additional sinusoidal channel (Eq. 14) and a corresponding GUI input; however, the OFAT tuning procedure itself scales linearly (or worse, given interactions) with the number of parameters, reinforcing the need for interaction-aware optimization methods as the parameter count grows. The baseline and OFAT comparisons used Webots' default (noise-free) IMU and joint-encoder models and default contact/friction parameters; the sensitivity of the reported RMSE and S/N results to realistic sensor noise and to contact-model uncertainty was not evaluated and is identified as a further necessary validation step prior to real-robot deployment.

D. STATISTICAL VALIDATION USING TAGUCHI SIGNAL-TO-NOISE (S/N) RATIO

While the comparative evaluation in Section IV-C demonstrates a visible reduction in CoM Roll and CoM Pitch tracking error under the OFAT-Optimized configuration, a formal statistical framework is required to determine whether this improvement constitutes a genuine reduction in dynamic "noise" rather than incidental run-to-run variation. To this end, the Taguchi method's Signal-to-Noise (S/N) ratio—a robust design metric widely used in quality and reliability engineering to quantify the sensitivity of a system response to uncontrolled disturbances—was employed to statistically validate the improvement achieved through the OFAT tuning process. In the context of robotic locomotion stability, tracking errors (RMSE) represent the "noise," i.e., the undesirable deviation of the robot's actual trajectory from its ideal reference path. Because smaller tracking errors are unambiguously preferable for both performance metrics under consideration, the *Smaller-the-Better* (STB) quality characteristic was selected for the S/N ratio calculation, in contrast to *Larger-the-Better* or *Nominal-the-Best* characteristics, which are not applicable to error-minimization objectives of this kind.

The S/N ratio (η) for a *Smaller-the-Better* response is mathematically defined as:

$$\eta = -10 \log_{10} \left(\frac{1}{n} \sum_{i=1}^{n} y_i^2 \right) \tag{16}$$

where y_i represents the measured error (in this case, the RMSE value) and n is the number of observations. Because we are evaluating the aggregate RMSE of the entire walking cycle, $n = 1$, simplifying the equation to $\eta = -10 \log_{10}(\text{RMSE}^2)$. Because $n = 1$ in this study (a single aggregate RMSE per configuration over the 11-second observation window), the S/N ratio reported here is a deterministic, logarithmic re-expression of the RMSE values already reported, rather than a statistical-significance measure computed across repeated trials. It should therefore be interpreted as a standardized, decibel-scale restatement of the RMSE improvement—useful for comparing the relative magnitude of the roll and pitch gains on a common scale—and not as evidence of trial-to-trial statistical robustness. Establishing statistical significance would require repeated trials under randomized initial or disturbance conditions ($n > 1$), which we identify as future work (Section V). By construction, the S/N ratio penalizes large errors logarithmically: a higher S/N ratio (less negative or higher positive, expressed in decibels) indicates a system response that is closer to the ideal target with minimal deviation, while a lower S/N ratio reflects greater sensitivity to disturbance and, correspondingly, weaker dynamic robustness.

Table 11 presents the comparative statistical analysis between the Default configuration and the OFAT-Optimized configuration, computed from the same oval-track trial data introduced in Section IV-C (Table 10), ensuring that the statistical validation is performed under the same continuously-turning locomotion condition used for the qualitative comparison.

TABLE 11: Statistical Comparison of Baseline vs. Optimal Configurations using Taguchi S/N Ratio (Oval Track)

Metric	Default		Optimized		$\Delta\eta$ (dB)
	RMSE	S/N	RMSE	S/N	
CoM Roll	0.1343	17.44	**0.1177**	**18.58**	$+1.14$
CoM Pitch	0.1894	14.45	**0.1713**	**15.32**	$+0.87$

The statistical evaluation in Table 11 confirms the qualitative trends observed in Figure 19, while additionally quantifying their significance on a logarithmic scale. The OFAT-Optimized configuration yields a $+1.14$ dB increase in the S/N ratio for lateral stability (CoM Roll) and a $+0.87$ dB increase for sagittal stability (CoM Pitch) relative to the Default configuration. Although these gains may appear numerically modest, it should be emphasized that the S/N ratio is a logarithmic measure: a $+1.14$ dB improvement corresponds to an approximate 12–13% reduction in mean-squared tracking error, which is consistent with the 12.4% RMSE reduction reported in Section IV-C. This confirms that the visible suppression of large CoM Roll and CoM Pitch excursions in Figures 19a and 19b reflects a statistically meaningful reduction in dynamic noise, rather than an artifact of the particular trial window selected. During the sequential OFAT stage (Tables 4–8), the widened foot clearance, extended double-support duration, and increased lateral swing amplitude each individually reduced roll and/or pitch RMSE relative to their respective baseline settings. We note, however, that Tables 4–8 characterize each parameter's isolated sensitivity during sequential tuning, not its marginal contribution within the final combined configuration evaluated on the oval track (Tables 10–11); a combinatorial ablation—reverting each parameter to its default value one at a time within the final combined configuration—was not performed and is identified as future work to more precisely decompose the 12.4%/9.6% RMSE reduction and the corresponding S/N gains among the five tuned parameters.

Overall, the Taguchi S/N ratio analysis provides a rigorous, single-number statistical confirmation that the proposed GUI-assisted OFAT tuning framework successfully minimized

kinematic "noise" in the robot's lateral and sagittal balance channels. This validates the practical efficacy of the interactive GUI not merely as a convenience tool for parameter adjustment, but as an effective instrument for achieving a statistically demonstrable improvement in bipedal walking stability.

Table 12 qualitatively compares the proposed sinusoidal-oscillator/GUI framework with the alternative gait-generation paradigms discussed in Section I, along dimensions relevant to the present study's engineering contribution rather than raw walking performance, which was not benchmarked on a common platform.

TABLE 12: Qualitative Comparison with Alternative Gait-Generation Paradigms

Approach	Strength	Cost / Limitation for Interactive Tuning
LIPM/ZMP [13], [15]	Formal balance guarantee	Requires accurate dynamic model and online ZMP computation
CPG [20], [43]	Adaptive inter-limb coordination	Oscillator-coupling design less directly exposed to human-in-the-loop tuning
MPC [23], [24]	Predictive disturbance rejection	High online computational overhead
Reinforcement Learning [21], [22]	Learned adaptability	Requires offline training and large data collection
Proposed (sinusoidal + GUI)	**Low computational overhead; real-time, human-in-the-loop tuning**	**No formal balance guarantee or online adaptability**

E. ADDITIONAL EVALUATION ON UNDULATING TERRAIN

The OFAT-Optimized configuration obtained in the preceding stages was further evaluated on an undulating circular track (Fig. 20). The terrain height was 10 mm from trough to crest (h10), with a sinusoidal amplitude of 5 mm and a wavelength of 0.60 m. A single native heightfield was simulated at an 8 ms time step. Terrain-contact settings were friction 1, restitution 0, soft CFM 10^{-5}, and a maximum of 10 contact joints; these differ from the default-contact settings used for the flat-track trials.

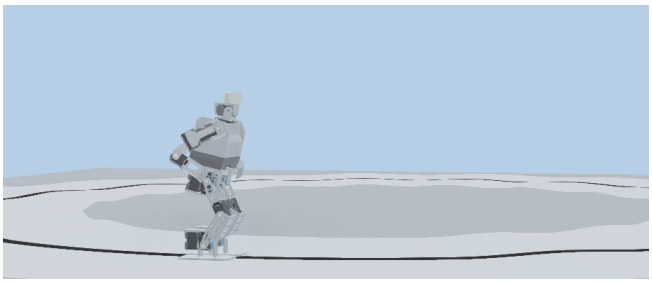
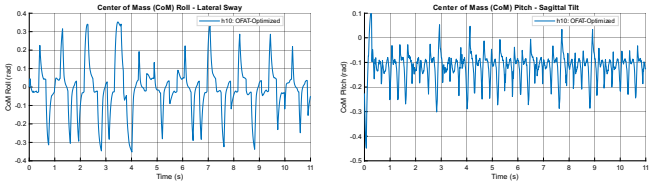


FIGURE 20: VoBiRo on the undulating circular track used for the additional h10 evaluation. The black line provides the visual path reference.



(a) (b)

FIGURE 21: Kinematic responses on the h10 undulating track over the first 11 s: (a) CoM Roll representing lateral sway and (b) CoM Pitch representing sagittal tilt.

Figure 21 presents CoM Roll and CoM Pitch over the first 11 s, using the same display window as the preceding comparisons. These terms follow the earlier sections and denote IMU-derived orientation angles. Full-record metrics were computed from all 13538 samples over 108.296 s, giving a mean GPS speed of 0.1756 m/s and zero-reference CoM Roll and CoM Pitch RMSE values of 0.1699 and 0.1345 rad, respectively. The mean pitch was -0.1118 rad, so its RMSE includes a sustained posture offset as well as oscillation.

No physical fall was observed during the recording. The logger's instantaneous acceleration-threshold flags were not treated as verified falls, and no samples were excluded. This trial extends the evaluation to the tested undulating-terrain and contact configuration. With only one h10 recording and no matched Default trial on the same terrain, the results do not establish comparative improvement or trial-to-trial robustness.

V. CONCLUSION

This paper presented the development of an interactive GUI-based framework for sinusoidal gait control and real-time simulation of the 22-DoF Vocational Bipedal Robot (VoBiRo) within the Webots environment. By establishing a non-blocking UDP communication channel between the Processing-based GUI and the Webots simulation, the proposed framework eliminated the conventional bottleneck of repeated hardcoding, recompilation, and physics engine restarts, enabling continuous and uninterrupted real-time tuning of gait parameters.

Using this framework, a systematic One-Factor-At-A-Time (OFAT) procedure was carried out to sequentially optimize five critical sinusoidal gait parameters—Period Time, Lateral Swing Amplitude, Foot Lift Height, Double Support Ratio, and Arm Swing Gain—based on the RMSE of CoM Roll and CoM Pitch. The resulting OFAT-Optimized configuration (600 ms period, 30 mm lateral swing, 45 mm foot lift, 16% DSR, and 2.0 arm swing gain) was then validated against the Default configuration on a more demanding closed-loop oval reference track, achieving a 12.4% reduction in CoM Roll RMSE and a 9.6% reduction in CoM Pitch RMSE. These improvements were further confirmed through the Taguchi Signal-to-Noise (S/N) ratio analysis, which yielded statistically meaningful gains of $+1.14$ dB for lateral stability and

+0.87 dB for sagittal stability, indicating that the observed reduction in tracking error reflects a genuine improvement in dynamic robustness rather than incidental trial-to-trial variation.

The additional 10 mm undulating-terrain trial sustained locomotion for 108.296 s with no visually observed fall, yielding CoM Roll and CoM Pitch RMSE values of 0.1699 and 0.1345 rad. This single-run result extends the tested operating conditions, but does not establish statistical robustness across terrain variations.

These findings confirm that the developed interactive GUI framework is an effective and efficient tool for accelerating gait parameter optimization and diagnosing multi-axis dynamic imbalances in bipedal locomotion, without requiring costly physical trials or repeated simulation restarts. Future work will focus on extending the framework toward semi-automated or optimization-driven tuning strategies, such as Taguchi orthogonal array design or metaheuristic search, to jointly optimize multiple gait parameters rather than sequentially, as well as validating the OFAT-Optimized configuration on the physical VoBiRo hardware to assess the transferability of the simulation-based results to real-world walking conditions.

We further note several limitations that define the scope of the present study and directions for future work. This study is limited to Webots simulation; the actuator model approximates the Dynamixel XL-430 servos through Webots' standard position-control dynamics and was not calibrated through dedicated system identification against the physical actuators (e.g., torque-speed curves, friction, backlash), and the baseline and OFAT comparisons used Webots' default contact and friction parameters without a systematic sensitivity sweep. The additional h10 evaluation used the terrain-contact settings specified in Section IV-E. Consequently, the sensitivity of the OFAT-optimized parameters to payload variation, terrain conditions beyond the single h10 evaluation, post-identification actuator dynamics, and simulator contact-parameter uncertainty, as well as the magnitude of the sim-to-real gap, remain unquantified. The stability assessment in this study is empirical, based on CoM roll/pitch RMSE and the Taguchi S/N ratio, and does not constitute a formal dynamical-systems stability guarantee; local orbital stability of the gait limit cycle (e.g., via Floquet multipliers or a Poincaré map), Lyapunov-based stability analysis, and estimation of the basin of attraction were not performed. Similarly, Zero-Moment-Point margin within the support polygon, capture-point tracking, and whole-body angular-momentum regulation were not monitored. Validating the OFAT-Optimized configuration on the physical VoBiRo hardware—including actuator system identification, formal limit-cycle stability analysis, and the complementary balance indicators above—is identified as essential future work to strengthen safety validation prior to physical deployment.

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